MIL-T-13499C
19 April 1988
SUPERSEDING
MIL-T-13499B
15 November 1973

MILITARY SPECIFICATION

TRAINING AID, MODEL BRIDGE:

SUPERSTRUCTURE, CLASS 60, PNEUMATIC

FLOAT, FLOATING BRIDGE: WITH CHEST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a model-bridge training aid disassembled and packed in a chest.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6910

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SPECIFICATIONS

FEDERAL

RR-C-271	- Chains and Attachments, Welded and Weldless.
DDD-T-86	- Tape, Textile; Cotton, General Purpose (Unbleached, Bleached or Dyed).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-R-3065	- Rubber, Fabricated Products.
MIL-C-12044	- Chests, Plywood.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-417	- Rubber Compositions, Vulcanized, General Purpose, Solid (Symbols and Tests).

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

D7891-1 to -30 inclusive	- Training Aid Bridge Model Steel Kit Superstructure-Class 60 Pneumatic Float Floating Bridge.
D7569-7	- Training Aid Bridge Model Kit M4 and M4A2 Floating Bridge Chain Hoist Details and Transverse Stiffener.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 3199 - Sponge, Chloroprene (CR) Rubber Firm.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The training aid and chest shall be as shown on Drawings D7891-1 through -30 and D7569-7 for the chain hoist details, and as specified herein. The components of the training aid shall be furnished disassembled and packed in a chest.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of training aids or training aids components from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnished training aids conforming to this specification.

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3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Wood. Wood for the abutment sill shall be cedar, basswood, ponderosa pine, sugar pine, white pine, yellow poplar, or redwood. Wood parts shall be surfaced-four-sides and shall be free from rot checks, worm holes, knots, or other defects.

3.3.2 Tape. The straps shall be fabricated from cotton tape conforming to DDD-T-86, 3/8-inch wide, type I, class 3, black.

3.3.3 Rubber. Tubes and rubber parts shall be fabricated from rubber compounds conforming to MIL-R-3065 and MIL-STD-417, polychloroprene, grade SC415 with C_2 suffix or grade SC420 with C_2 suffix, or butyl, grade RS420 with A_1 and C_2 suffix. Sponge rubber parts shall be fabricated from rubber compounds conforming to SAE AMS 3199.

3.3.4 Chain. The chain shall conform to RR-C-271, type II, class 7, brass, 0.048-inch diameter wire.

3.4 Tubes. The rubber tubes shall withstand an internal pressure of 1.5 pounds per square inch (psi) for a period of 4 hours without leakage.

3.5 Chests. Chests shall conform to the applicable requirements of MIL-C-12044.

3.6 Treatment and painting. The portions of the training aid and chest and their components and parts normally painted in commercial practice shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.7 Assembly. The training aid and chest shall be assembled without forcing, bending, or springing any of the components.

3.8 Workmanship. The training aid and chest shall be free from defects that could adversely affect their assembly and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article training aid and chest shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article training aid shall be tested as specified in 4.5.2. Failure to pass any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.4.1.2 Tests. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective.

4.4.3 Tests.

4.4.3.1 Individual. Each rubber tube shall be tested as specified in 4.5.2.2. Failure of the test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.5.2.1. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

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4.5.1 Examination. The training aid and chest shall be examined for the following defects:

101. Dimensions not as specified.
102. Materials not as specified.
103. Components missing or fabrication not as specified.
104. Treatment and painting not as specified.
105. Workmanship not as specified.
106. Rubber components torn or not securely vulcanized.
107. Loading plan not as shown on drawings.

4.5.2 Tests.

4.5.2.1 Assembly. Assemble the training aid. Failure of the training aid to assemble without forcing, twisting, or springing of components shall constitute failure of this test.

4.5.2.2 Inflation. Inflate the rubber tubes to a pressure of 1.5 psi. Any leakage or deflation within a period of 4 hours shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. Each packaged training aid shall be examined for the following defects. Any defects found shall be corrected prior to shipment.

108. Pump not preserved as specified for level A.
109. Tools not preserved and packaged in accordance with the referenced document.
110. Components not cushioned to prevent movement or damage as specified for level A.
111. Packing not as specified for level A, B, or C.
112. Marking missing, illegible, incorrect, or incomplete for level A, B, or C.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Pumps. Each pump shall be preserved in accordance with MIL-P-116, method IC-1.

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5.1.1.2 Tools. The tools shall be preserved in accordance with the level A requirements of PPP-P-40.

5.1.1.3 Screws, nuts, and pins. The screws, nuts, and pins shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.4 Consolidated packaging. The components comprising a complete training aid shall be consolidated in the chests in accordance with the chest loading plans. Cushioning material conforming PPP-C-843, type I, class A or other appropriate, resilient cushioning material shall be used to prevent movement of contents within the chests. Chest lids shall be closed and securely latched.

5.1.2 Level C. The components comprising a complete training aid, contained in the chests in accordance with the chest loading plan, shall be preserved to afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The chests containing the components comprising a complete training aid shall be packed together in a close fitting box conforming to PPP-B-601, overseas type, grade B, style optional or PPP-B-621, class 2, style 2. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall be flat with finish B.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except boxes shall be domestic type or class as applicable, and strapping shall be finish A.

5.2.3 Level C. The chests containing one complete training aid shall be palletized or shall be consolidated in a single, sturdy container having strength characteristics appropriate for the size and weight of the chests contained. Chests shall be blocked to prevent movement within the container. If palletized, the chests shall be secured to the pallet with flat steel strapping.

5.3 Marking. Chests and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The model-bridge training aid is intended as an instructional aid for training of troops in the erection of the prototype.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is not required for inspection and approval, and the number of units required (see 3.2).
c. Level of preservation and level of packing required (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the items should be an initial production item. The first article should consist of one or more units. The contracting office should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Subject term (key word).

Class 60
Floating bridge
M4
M4A2
Model bridge
Pneumatic float
Training aid

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 70

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 6910-0012

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-13499C	Training Aid, Model Bridge: Superstructure, Class 60, Pneumatic Float, Floating Bridge: With Chest

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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